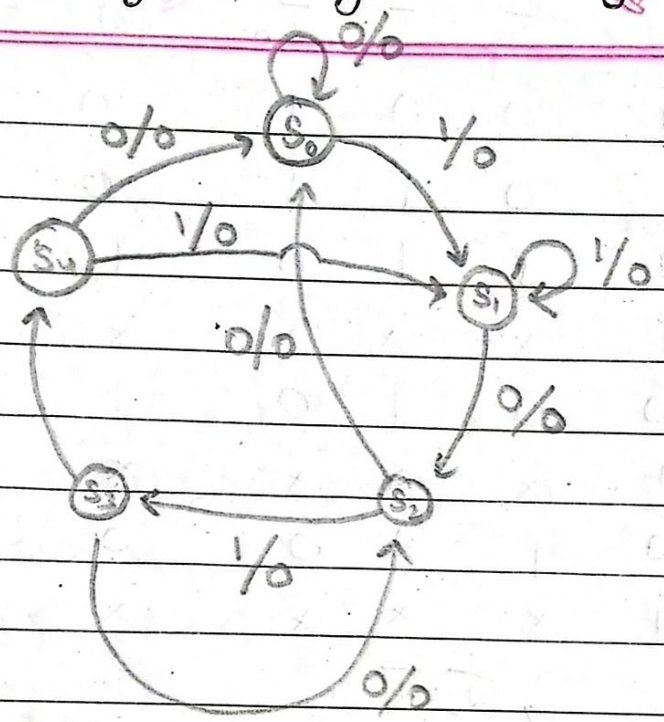
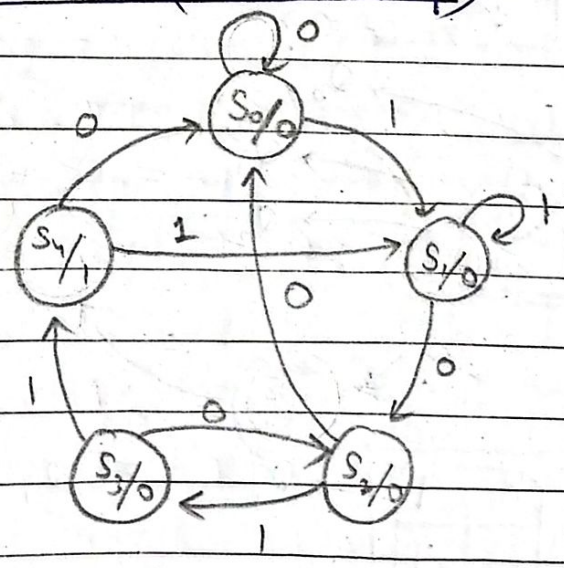
1. Draw the state diagram to detect the sequence "1011" using both mealy and more machine. Implement the same using Verilog and verify it using test bench.

Mealy ~~Moore~~ Machine (Non-Overlap)



Moore ~~Mealy~~ Machine (Non-Overlap)



VERILOG MODULE

```
`timescale 1ns / 1 ps
module seq_detector(
    input x, clk, reset,
    output reg z
);
```

```
parameter S0=0 , S1=1 , S2=2 , S3=3 , S4=4;
reg [1:0] PS, NS;

always @ (posedge clk or posedge reset)
    begin
        if (reset)
            PS <= S0;
        else
            PS <= NS;
    end

always @ (PS or x)
    begin
        case (PS)
            S0 : begin
                    z=0;
                    NS = x ? S1 : S0;
                    $display (PS);
                 end
            S1 : begin
                    z=0;
                    NS= x ? S1 : S2;
                    $display (PS);
                 end
            S2 : begin
                    z=0;
                    NS= x ? S3 : S0;
                    $display (PS);
                 end
            S3 : begin
                    z=0;
                    NS = x ? S4 : S2;
                    $display (PS);
                 end
```

```
S4: begin
        z=1;
        NS = PS? S1 : S0;
        $display(PS);
    end
  endcase
end
end module
```

TEST Bench

```
`timescale 1ns / 1ps
module testbench;
    // Inputs
    reg x;
    reg clk;
    reg reset;
    // OP
    wire z;
    seq_detector uut (
        .x(x), .clk(clk), .reset(reset),
        .z(z) );
initial
    begin
        clk = 1'b0;
        reset = 1'b1;
        #15 reset = 1'b0;
    end
always #5 clk = ~clk;
initial begin
    #12 x=0; #10 x=0; #10 x=1; #10 x=0;
    #12 x=1; #10 x=1; #10 x=0; #10 x=1;
    #12 x=1; #10 x=0; #10 x=0; #10 x=1;
    #12 x=0; #10 x=1; #10 x=1; #10 x=0;
    #10 $finish;
end
end module
```